Experiment No. 4

Title

Searching for Files and Text using grep and related in Linux

Aim

To understand and perform searching of text patterns and files in Linux using `grep` and other related commands.

Theory

Text Searching in Linux

Linux treats most data as plain text, making text-processing tools extremely powerful. Searching text efficiently is essential for:

- Log file analysis
- Configuration troubleshooting
- Security monitoring
- Automation scripts

grep Command

The **grep** command searches for a pattern in files or input streams and prints matching lines.

Basic syntax:

grep [options] pattern filename

Related Commands

- grep – General pattern searching
- fgrep – Fast grep (searches fixed strings, no regular expressions)
- egrep – Extended grep (supports extended regular expressions)

Note: In modern Linux systems, egrep and fgrep are implemented using grep -E and grep -F.

Commands Used

grep	Search text using patterns
grep -i	Case-insensitive search
grep -n	Show line numbers
grep -r	Recursive search
grep -v	Invert match

fgrep	Search fixed strings
cat	Display file content
ls	List files

Procedure / Steps

• Log in to the Linux system and open the terminal.

• Create a working directory:

```
mkdir grep_lab
cd grep_lab
```

• Create a sample text file:

```
nano sample.txt
```

• Add content to files using redirection:

```
Linux is powerful
Linux is secure
Bash scripting is useful
Cyber Security uses Linux
```

• Display file contents:

```
cat file1.txt
```

• Search for a word using grep:

```
grep Linux sample.txt
```

• Perform case-insensitive search:

```
grep -i linux sample.txt
```

• Display line numbers with matches:

```
grep -n Linux sample.txt
```

• Invert the match:

```
grep -v Linux sample.txt
```

• Search using fgrep:

```
fgrep "Bash scripting" sample.txt
```

• Perform recursive search (if multiple files exist):

```
grep -r Linux .
```

Sample Execution

•

```
┌──(setika㉿kali)-[~/grep_lab]
└─$ grep -i linux sample.txt
linux is powerful
linux is secure
cyber security uses linux

┌──(setika㉿kali)-[~/grep_lab]
└─$ grep -n Linux sample.txt

┌──(setika㉿kali)-[~/grep_lab]
└─$ grep -v Linux sample.txt
linux is powerful
linux is secure
bash scripting is useful
cyber security uses linux
```

Outcomes

• Matching lines containing the search pattern are displayed.
• Line numbers are shown where applicable.
• Case-insensitive and inverted searches work correctly.

Result

Text searching operations were successfully performed using grep and related commands. The student gained practical experience in locating specific patterns within files, which is essential for log analysis and automation tasks.